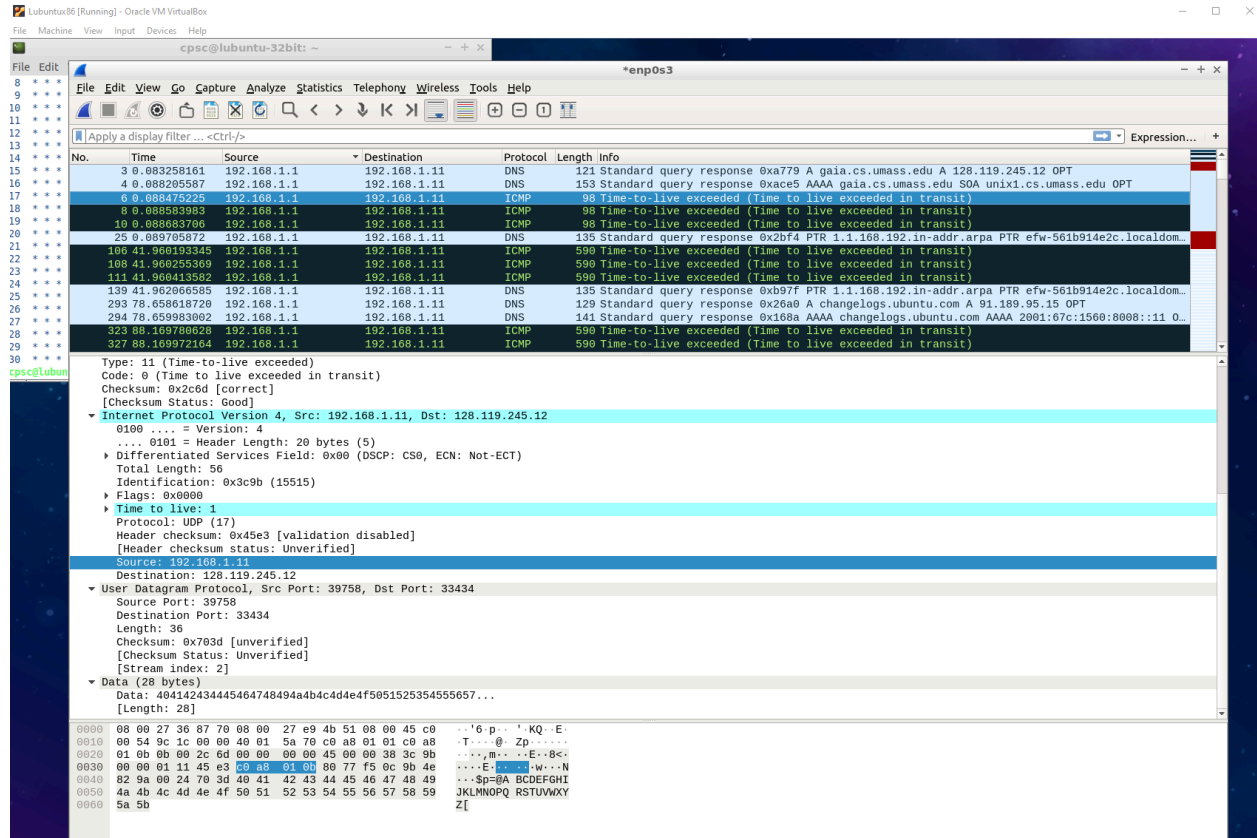


Lab 3: IPv4

1. Select the first ICMP Echo Request message sent by your computer, and expand the Internet Protocol part of the packet in the packet details window. What is the IP address of your computer?

Since I ran this test using the lubuntu VM, this is the screen shot provided.



The IP address of the VM is 192.168.1.11

2. Within the IP packet header, what is the value in the upper layer protocol field?

The value in the protocol field is ICMP (1)

6 0.088475225	192.168.1.1	192.168.1.11	ICMP	98 Tim
8 0.088583983	192.168.1.1	192.168.1.11	ICMP	98 Tim
10 0.088683706	192.168.1.1	192.168.1.11	ICMP	98 Tim
25 0.089705872	192.168.1.1	192.168.1.11	DNS	135 Sta
106 41.960193345	192.168.1.1	192.168.1.11	ICMP	590 Tim
108 41.960255369	192.168.1.1	192.168.1.11	ICMP	590 Tim
111 41.960413582	192.168.1.1	192.168.1.11	ICMP	590 Tim
139 41.962066585	192.168.1.1	192.168.1.11	DNS	135 Sta
293 78.658618720	192.168.1.1	192.168.1.11	DNS	129 Sta
294 78.659983002	192.168.1.1	192.168.1.11	DNS	141 Sta
323 88.169780628	192.168.1.1	192.168.1.11	ICMP	590 Tim
327 88.169972164	192.168.1.1	192.168.1.11	ICMP	590 Tim

▶ Frame 6: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface
▶ Ethernet II, Src: PcsCompu_e9:4b:51 (08:00:27:e9:4b:51), Dst: PcsCompu_36:87:70
▼ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.11
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 ▶ Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
 Total Length: 84
 Identification: 0x9c1c (39964)
 ▶ Flags: 0x0000
 Time to live: 64
 Protocol: ICMP (1)
 Header checksum: 0x5a70 [validation disabled]
 [Header checksum status: Unverified]
 Source: 192.168.1.1
 Destination: 192.168.1.11
▼ Internet Control Message Protocol

3. How many bytes are in the IP header? How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes.

▼ Internet Protocol Version 4, Src: 192.168.1.11, Dst: 128.119.245.12

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 56

Identification: 0x3c9b (15515)

▶ Flags: 0x0000

▶ Time to live: 1

Protocol: UDP (17)

Header checksum: 0x45e3 [validation disabled]

[Header checksum status: Unverified]

Source: 192.168.1.11

Destination: 128.119.245.12

▶ User Datagram Protocol, Src Port: 39758, Dst Port: 33434

▶ Data (28 bytes)

The total length is 56 bytes, and the header is 20 bytes, so the payload of the datagram is 36 bytes.

(56 total – 20 for header = 36 bytes)

4. Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented.

```
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 56
  Identification: 0x3c9b (15515)
▼ Flags: 0x0000
  0... .... = Reserved bit: Not set
  .0.. .... = Don't fragment: Not set
  ..0. .... = More fragments: Not set
  ...0 0000 0000 0000 = Fragment offset: 0
▶ Time to live: 1
Protocol: UDP (17)
```

It has not been fragmented, otherwise the more fragments flag would be set.

5. Which fields in the IP datagram always change from one datagram to the next within this series of ICMP messages sent by your computer?

```

Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.11
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0xc0 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x9c1c (39964)
    Flags: 0x0000
    Time to live: 64
Protocol: ICMP (1)
  Header checksum: 0x5a70 [validation disabled]
  [Header checksum status: Unverified]
  Source: 192.168.1.1
  Destination: 192.168.1.11
Internet Control Message Protocol
  Type: 11 (Time-to-live exceeded)
  Code: 0 (Time to live exceeded in transit)
  Checksum: 0x2c6d [correct]
  [Checksum Status: Good]
Internet Protocol Version 4, Src: 192.168.1.11, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 56
    Identification: 0x3c9b (15515)
    Flags: 0x0000
    Time to live: 1
    Protocol: UDP (17)
    Header checksum: 0x45e3 [validation disabled]
    [Header checksum status: Unverified]
    Source: 192.168.1.11
    Destination: 128.119.245.12
User Datagram Protocol, Src Port: 39758, Dst Port: 33434
  Source Port: 39758
  Destination Port: 33434

Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.11
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0xc0 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x9c1d (39965)
    Flags: 0x0000
    Time to live: 64
Protocol: ICMP (1)
  Header checksum: 0x5aef [validation disabled]
  [Header checksum status: Unverified]
  Source: 192.168.1.1
  Destination: 192.168.1.11
Internet Control Message Protocol
  Type: 11 (Time-to-live exceeded)
  Code: 0 (Time to live exceeded in transit)
  Checksum: 0x2c6d [correct]
  [Checksum Status: Good]
Internet Protocol Version 4, Src: 192.168.1.11, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 56
    Identification: 0x3c9c (15516)
    Flags: 0x0000
    Time to live: 1
    Protocol: UDP (17)
    Header checksum: 0x45e2 [validation disabled]
    [Header checksum status: Unverified]
    Source: 192.168.1.11
    Destination: 128.119.245.12
User Datagram Protocol, Src Port: 43357, Dst Port: 33435
  Source Port: 43357
  Destination Port: 33435

Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.11
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0xc0 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x9c1e (39966)
    Flags: 0x0000
    Time to live: 64
Protocol: ICMP (1)
  Header checksum: 0x5a8e [validation disabled]
  [Header checksum status: Unverified]
  Source: 192.168.1.1
  Destination: 192.168.1.11
Internet Control Message Protocol
  Type: 11 (Time-to-live exceeded)
  Code: 0 (Time to live exceeded in transit)
  Checksum: 0x2c6d [correct]
  [Checksum Status: Good]
Internet Protocol Version 4, Src: 192.168.1.11, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 56
    Identification: 0x3c9d (15517)
    Flags: 0x0000
    Time to live: 1
    Protocol: UDP (17)
    Header checksum: 0x45e1 [validation disabled]
    [Header checksum status: Unverified]
    Source: 192.168.1.11
    Destination: 128.119.245.12
User Datagram Protocol, Src Port: 36346, Dst Port: 33436
  Source Port: 36346
  Destination Port: 33436

```

In these series of ICMP messages, the Identification field and the Header checksum always change. One thing I noticed in the screenshots provided by the author, the TTL increments but it does not for mine.

6. Which fields stay constant? Which of the fields must stay constant? Which fields must change? Why?

The version is always IPv4 so it'll stay the same, header length is also the same this reason so they must stay constant. Source IP and destination will stay constant because my IP and the IP of the website won't change. The Differentiated Services field is also the same since the protocol does not change. For that matter, the Protocol field does not change and is thus constant. The flags also remain constant since it's not fragmented.

In my attempt for the lab, the TTL stayed constant. For reference I was using the lubuntu VM. The exact commands I used were 'tracert gaia.cs.umass.edu 56' then 2000 then 3500. In the screenshots, the author seems to be mostly using a windows environment so I'll assume this just has to do with the way I performed my test, since I believe tracert should increase the TTL for each hop.

The fields that must change are the Identification fields, and possibly TTL, but this was not the case in my situation.

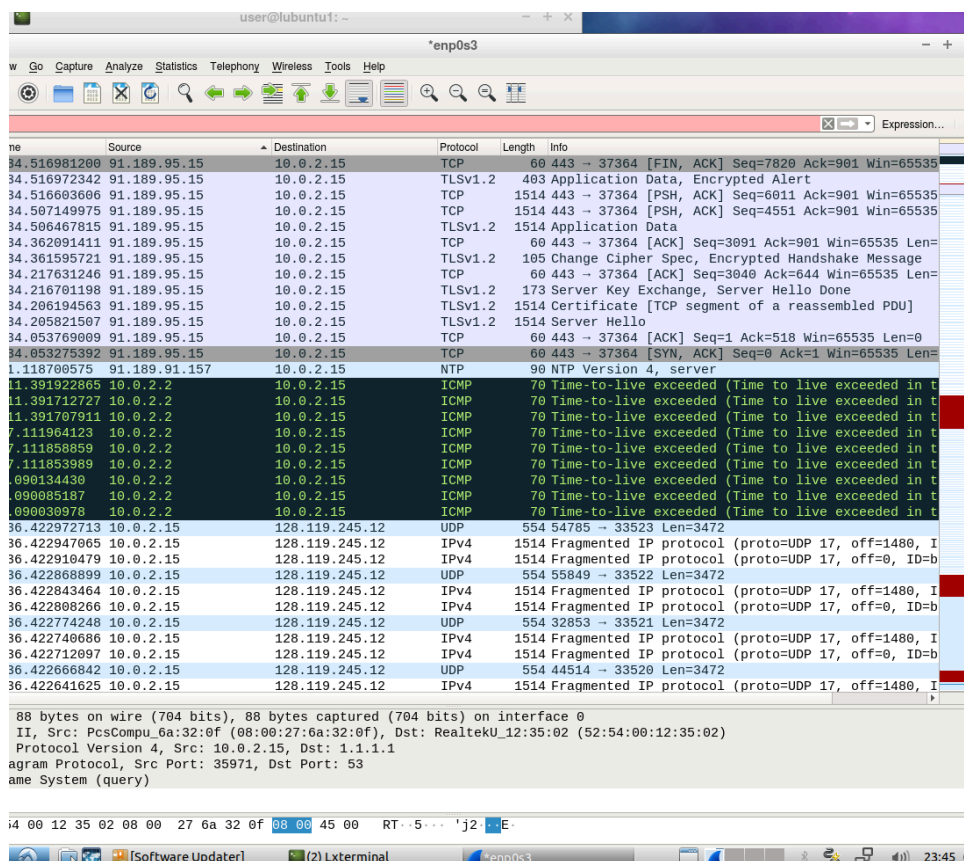
7. Describe the pattern you see in the values in the Identification field of the IP datagram

They increment by one each time.

Next (with the packets still sorted by source address) find the series of ICMP TTL-exceeded replies sent to your computer by the nearest (first hop) router.

8. What is the value in the Identification field and the TTL field?

In my case, I was getting confused because my screenshots were not matching up with what was provided by the author. I attempted to run this test again without using the provided VM's and a fresh ubuntu ISO file. This did not change anything, as can be seen below.



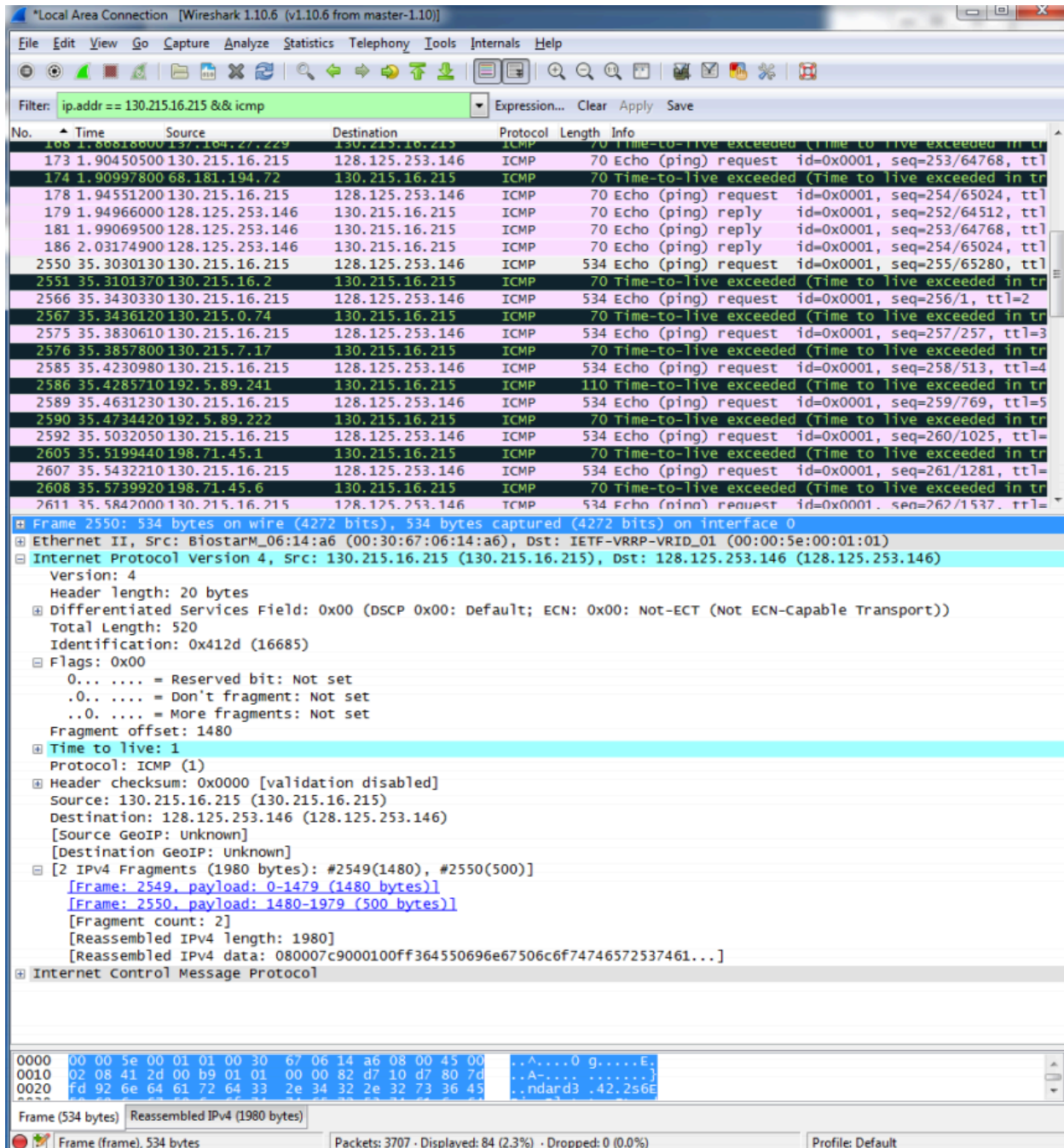
This results in the same 9 TTLexceeded packets as when I ran it on the provided VM's. So I went back to using the provided VM's to solve this problem (Screenshots further above). I could not see any replies sent back to my computer, I am assuming due to the fact that the author used a windows environment. Either way, the first traceroute was for 56 bytes, the second for 2000, and the final one for 3500 bytes. The TTL is 64 for each one.

9. Do these values remain unchanged for all of the ICMP TTL-exceeded replies sent to your computer by the nearest (first hop) router? Why?

These values do not change. I'm not sure exactly why, as I believe the TTL should be increasing from the author's screenshots provided in the book. I have spent more than an hour trying to trouble shoot these issues, but I presume it's just because the first hop router is my home router or something.

10. Find the first ICMP Echo Request message that was sent by your computer after you changed the Packet Size in pingplotter to be 2000. Has that message been fragmented across more than one IP datagram? [Note: if you find your packet has not been fragmented, you should download the zip file <http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip> and extract the ipethereal-trace-1packet trace. If your computer has an Ethernet interface, a packet size of 2000 should cause fragmentation.³]

In my case, I used provided screenshots, as mine was not fragmented.



This message HAS been fragmented, as can be seen by Frame 2549 and 2550

11. Take a screenshot of the first fragment of the fragmented IP datagram. What information in the IP header indicates that the datagram been fragmented? What information in the IP header indicates whether this is the first fragment versus a latter fragment? How long is this IP datagram?

The more fragments flag being set shows that the datagram is fragmented. The fragment offset is 0 so it's the first fragment. The length of the datagram is 1500.

Local Area Connection [Wireshark 1.10.6 (v1.10.6 from master-1.10)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: ip.addr==130.215.16.215

No.	Time	Source	Destination	Protocol	Length	Info
2549	35.3029980	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=255/65280, ttl=3
2550	35.3030130	130.215.16.215	128.125.253.146	IPv4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=255)
2551	35.3101370	130.215.16.2	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2565	35.3430210	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=256/1, ttl=2
2566	35.3430330	130.215.16.215	128.125.253.146	IPv4	534	Fragmented IP protocol (proto=ICMP 1, off=2480, ID=256)
2567	35.3436120	130.215.0.74	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2574	35.3830450	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=257/257, ttl=3
2575	35.3830610	130.215.16.215	128.125.253.146	IPv4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=257)
2576	35.3831260	130.215.16.215	128.125.253.146	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2584	35.4230820	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=258/513, ttl=4
2585	35.4230980	130.215.16.215	128.125.253.146	IPv4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=258)
2586	35.4285710	192.5.89.241	130.215.16.215	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
2588	35.4631120	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=259/769, ttl=5
2589	35.4631230	130.215.16.215	128.125.253.146	IPv4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=259)
2590	35.4734420	192.5.89.222	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2591	35.5031940	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=260/1025, ttl=6
2592	35.5032050	130.215.16.215	128.125.253.146	IPv4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=260)
2606	35.5431920	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=261/1281, ttl=1
2607	35.5432210	130.215.16.215	128.125.253.146	IPv4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=261)
2608	35.5739920	198.71.45.6	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)

2549: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface 0

Ethernet II, Src: BiostarM_06:14:a6 (00:30:67:06:14:a6), Dst: IETF-VRRP-VRID_01 (00:00:5e:00:01:01)

Internet Protocol Version 4, Src: 130.215.16.215 (130.215.16.215), Dst: 128.125.253.146 (128.125.253.146)

Version: 4

Header Length: 20 bytes

Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECT-Capable Transport))

Total Length: 1500

Identification: 0x412d (16685)

Flags: 0x01 (More Fragments)

0... .. = Reserved bit: Not set

.0... .. = Don't fragment: Not set

.1... .. = More fragments: Set

Fragment offset: 0

Time to live: 1

Protocol: ICMP (1)

Header checksum: 0x0000 [validation disabled]

Source: 130.215.16.215 (130.215.16.215)

Destination: 128.125.253.146 (128.125.253.146)

[Source GeoIP: unknown]

[Destination GeoIP: unknown]

Internet Control Message Protocol

0000 00 00 5e 00 01 01 00 30 67 06 14 a6 08 00 45 00 ..^....0 g....E.

0010 05 dc 41 2d 20 00 01 01 00 00 82 d7 10 d7 80 7d ..A-... ..}.

0020 fd 92 08 00 07 c9 00 01 00 ff 36 45 50 69 6e 676ePing

0030 50 6c 6f 74 74 65 72 53 74 61 6e 64 61 72 64 33 42 2566 PingPlot

0040 2e 34 32 2e 32 73 36 45 50 69 6e 67 50 6c 6f 74 .42.2566 PingPlot

0050 74 65 72 53 74 61 6e 64 61 72 64 33 2e 34 32 2e terstand ard3.42.

0060 32 73 36 45 50 69 6e 67 50 6c 6f 74 74 65 72 53 2566 Ping Plotters

0070 74 61 6e 64 61 72 64 33 2e 34 32 2e 32 73 36 45 tandard3 .42.2566

File: C:\Users\drl\AppData\Local\Temp\wi... Packets: 3707 - Displayed: 148 (4.0%) - Dropped: 0 (0.0%)

Profile Default

12. Print out the second fragment of the fragmented IP datagram. What information in the IP header indicates that this is not the first datagram fragment? Are there more fragments? How can you tell?

The screenshot shows a Wireshark 1.10.6 interface with a packet capture of an ICMP Echo request. The packet list shows a fragmented IP protocol (proto=ICMP 1, off=1480, ID=255) at packet 2550. The packet details pane shows the IP header with 'Fragment offset: 1480' and 'More fragments: Not set'. The packet bytes pane shows the raw data of the fragmented packet.

No.	Time	Source	Destination	Protocol	Length	Info
2549	35.3029980	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=255/65280, ttl=3
2550	35.3030130	130.215.16.215	128.125.253.146	IPV4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=255)
2551	35.3101370	130.215.16.215	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2565	35.3430210	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=256/1, ttl=2
2566	35.3430330	130.215.16.215	128.125.253.146	IPV4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=256)
2567	35.3436120	130.215.16.215	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2574	35.3830450	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=257/257, ttl=3
2575	35.3830610	130.215.16.215	128.125.253.146	IPV4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=257)
2576	35.3857800	130.215.16.215	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2584	35.4230820	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=258/513, ttl=4
2585	35.4230980	130.215.16.215	128.125.253.146	IPV4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=258)
2586	35.4285710	192.5.89.241	130.215.16.215	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
2588	35.4631120	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=259/769, ttl=5
2589	35.4631230	130.215.16.215	128.125.253.146	IPV4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=259)
2590	35.4734420	192.5.89.222	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2591	35.5031940	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=260/1025, ttl=6
2592	35.5032050	130.215.16.215	128.125.253.146	IPV4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=260)
2605	35.5199440	198.71.45.1	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
2606	35.5431920	130.215.16.215	128.125.253.146	ICMP	1514	Echo (ping) request id=0x0001, seq=261/1281, ttl=6
2607	35.5432210	130.215.16.215	128.125.253.146	IPV4	534	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=261)
2608	35.5739920	198.71.45.6	130.215.16.215	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)

Frame 2550: 534 bytes on wire (4272 bits), 534 bytes captured (4272 bits) on interface 0

Ethernet II, Src: BiostarM_06:14:a6 (00:30:67:06:14:a6), Dst: IETF-VRRP-VRID_01 (00:00:5e:00:01:01)

Internet Protocol Version 4, Src: 130.215.16.215 (130.215.16.215), Dst: 128.125.253.146 (128.125.253.146)

Version: 4

Header length: 20 bytes

Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))

Total Length: 520

Identification: 0x412d (16685)

Flags: 0x00

0... = Reserved bit: Not set

.0.. = Don't fragment: Not set

..0. = More fragments: Not set

Fragment offset: 1480

Time to live: 1

Protocol: ICMP (1)

Header checksum: 0x0000 [validation disabled]

Source: 130.215.16.215 (130.215.16.215)

Destination: 128.125.253.146 (128.125.253.146)

[Source GeoIP: Unknown]

[Destination GeoIP: Unknown]

Data (500 bytes)

0000 00 00 5e 00 01 01 00 30 67 06 14 a6 08 00 45 00 ..^....0 g.....E.

0010 02 08 41 2d 00 b9 01 01 00 00 82 d7 10 d7 80 7d ..A-....

0020 fd 92 6e 64 61 72 64 33 2e 34 32 2e 32 73 36 45 ..ndard3 .42.2s6E

0030 50 69 6e 67 50 6c 6f 74 74 65 72 53 74 61 6e 64 PingPlot terStand

0040 61 72 64 33 2e 34 32 2e 32 73 36 45 50 69 6e 67 ard3.42. 2s6EPing

0050 50 6c 6f 74 74 65 72 53 74 61 6e 64 61 72 64 33 PlotterS tandard3

0060 2e 34 32 2e 32 73 36 45 50 69 6e 67 50 6c 6f 74 .42.2s6E PingPlot

0070 74 65 72 53 74 61 6e 64 61 72 64 33 2e 34 32 2e terStand ard3.42.

0080 32 73 36 45 50 69 6e 67 50 6c 6f 74 74 65 72 53 2s6EPing PlotterS

In the screenshot above, you can see that the offset is 1480 and since the “More Fragments” flag is not set, it is the last fragment.

13. What fields change in the IP header between the first and second fragment?

The total length, the more fragments bit, and the offset, and checksum.

Now find the first ICMP Echo Request message that was sent by your computer after you changed the Packet Size in pingplotter to be 3500.

The screenshot shows a Wireshark packet capture on the 'Local Area Connection' interface. The filter is set to 'ip.addr == 130.215.16.215'. The packet list shows several ICMP Echo (ping) requests and replies, as well as fragmented IP protocols. The packet details pane is expanded for packet 3567, showing the following fields:

- Frame 3567: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface 0
- Ethernet II, Src: Biostarm_06:14:a6 (00:30:67:06:14:a6), Dst: IETF-VRRP-VRID_01 (00:00:5e:00:01:01)
- Internet Protocol Version 4, Src: 130.215.16.215 (130.215.16.215), Dst: 128.125.253.146 (128.125.253.146)
 - Version: 4
 - Header length: 20 bytes
 - Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
 - Total Length: 1500
 - Identification: 0x413d (16701)
 - Flags: 0x01 (More Fragments)
 - 0... = Reserved bit: Not set
 - .0.. = Don't fragment: Not set
 - ..1. = More fragments: Set
 - Fragment offset: 0
 - Time to live: 1
 - Protocol: ICMP (1)
 - Header checksum: 0x0000 [validation disabled]
 - Source: 130.215.16.215 (130.215.16.215)
 - Destination: 128.125.253.146 (128.125.253.146)
 - [Source GeoIP: Unknown]
 - [Destination GeoIP: unknown]
- Internet Control Message Protocol

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, the IPv4 header, and the ICMP Echo (ping) request data.

14. How many fragments were created from the original datagram?

There were 3 fragments created according to the provided screenshots.

15. What fields change in the IP header among the fragments?

The fragment offset, checksum. In the first two packets, the more fragments flag is set and the total length is 1500. The third's packet total length is 540, and more fragments is not set.